Final Project Report

AI-Driven Emotion Detection & Personalized Learning Support Platform

Abstract

The AI-Driven Emotion Detection & Personalized Learning Support Platform is an intelligent educational system designed to detect learners' emotions from text input and provide personalized learning guidance. The system combines BERT and BiLSTM deep learning models with Google Gemini AI to understand learner emotions such as Confused, Curious, Frustrated, Bored, and Confident. Based on the detected emotion, the platform generates supportive responses and learning suggestions. The application is developed using Python and Streamlit, providing an interactive web interface. It also stores session history for future analysis and helps improve learner engagement.

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1. Introduction

Online learning has become an essential part of education. However, most learning platforms do not understand the emotional state of learners. Students often experience confusion, frustration, or boredom while studying, reducing their learning effectiveness.

This project addresses this challenge by integrating Artificial Intelligence with Emotion Detection and Generative AI to provide personalized learning support.

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2. Problem Statement

Traditional e-learning platforms provide identical content to every learner regardless of their emotional state. Students who require personalized guidance often do not receive immediate assistance.

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3. Objectives

- Detect learner emotions from text.
- Generate personalized AI guidance.
- Improve student engagement.
- Reduce frustration during learning.
- Provide a user-friendly AI learning assistant.

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4. Technologies Used

- Python
- Streamlit
- TensorFlow
- Hugging Face Transformers
- BERT
- BiLSTM
- Google Gemini API
- Pandas
- NumPy

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5. System Architecture

User Input

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Text Preprocessing

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Emotion Detection (BERT + BiLSTM)

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Google Gemini AI

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Personalized Learning Suggestion

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Session History Storage

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6. Modules

Module 1

User Details

Module 2

Emotion Detection

Module 3

Gemini AI Response

Module 4

Session History

Module 5

Dashboard

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7. Results

The application successfully predicts learner emotions and generates personalized AI-based learning guidance. The system provides fast responses through a simple Streamlit interface.

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8. Future Scope

- Voice-based emotion detection
- Multi-language support
- Mobile application
- Teacher analytics dashboard
- Cloud deployment

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9. Conclusion

The project successfully integrates Deep Learning and Generative AI to create an intelligent learning assistant that improves learner engagement through emotion-aware personalized support.

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10. References

- Google Gemini API Documentation
- Hugging Face Transformers Documentation
- TensorFlow Documentation
- Streamlit Documentation